

03 PROTOCOL

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Course: Research Methods and Scientific Integrity in AI and Advanced Technologies

Program: Doctoral Program in Deep Technologies – UNMSM

1. TITLE

"Intelligent Digital Twin Model with Cognitive Learning for Prediction and Optimization of Costs, Schedules, and Risks in Construction Projects"

Short title: Cognitive Digital Twin for Construction Optimization

2. ABSTRACT

The Peruvian construction industry faces a chronic crisis: 87% of building projects in Metropolitan Lima deviate more than 15% from their original budget, and 76% exceed contractual deadlines (CAPECO, 2025). This research develops an intelligent digital twin model with cognitive learning that, using historical project data, predicts cost overruns, schedule delays, and risks during project execution. The research adopts a design science paradigm (building the DT-ML artifact) combined with a quantitative approach to evaluate its predictive accuracy. The methodology includes four phases: (1) structuring historical data from 40-100 construction projects in Peru, (2) designing the digital twin architecture with four layers (data capture, virtual model, ML engine, visualization), (3) training and comparing cognitive learning algorithms (CatBoost, XGBoost, Random Forest, LSTM, Case-Based Reasoning), and (4) validating the model in real projects compared to traditional methods (CPM, EVM). Expected outcomes include reducing cost overrun prediction error from the current 15-20% (expert judgment) to less than 10% (MAPE), and reducing budget deviations by at least 30%. The model will be accessible to Peruvian medium-sized construction companies (implementation cost < \$15,000 USD), closing the technology gap that currently excludes most construction firms in the country.

3. INTRODUCTION AND PROBLEM STATEMENT

3.1 Current Situation

The construction industry in Peru is the backbone of the country's infrastructure development, yet it operates with a level of inefficiency that would be unacceptable in any other sector. According to the Peruvian Chamber of Construction (CAPECO, 2025), 87% of building projects in Metropolitan Lima present deviations greater than 15% from their original budget, and 76% exceed contractual deadlines. Globally, the situation is equally alarming: 98% of infrastructure projects suffer cost overruns or delays, with an average cost overrun of 80% above the original budget (Flyvbjerg, 2024). These losses translate into approximately \$1.5 trillion annually worldwide.

3.2 The Specific Problem

Peruvian construction companies cannot anticipate cost and schedule deviations for three main reasons:

First barrier: lack of digitalized institutional memory. Historical data is scattered across Excel files, scanned PDFs, and MS Project files, without a homogeneous format that enables systematic analysis. Lessons learned from past projects—specifically regarding actual versus planned costs and activity durations—remain stored in disconnected files or in the memory of professionals who leave the company.

Second barrier: absence of a predictive culture. The majority of Peruvian construction professionals rely on expert judgment ("the site supervisor's experience") over statistical models. A recent qualitative study found that 68% of resident engineers in Lima consider that "historical data does not reflect the unique conditions of each project" (Paredes & Quispe, 2024)—a belief that, although partially true, has hindered investment in analytical solutions.

Third barrier: high costs of commercial solutions. Commercial Digital Twin solutions (such as those from Bentley Systems, Autodesk Tandem, or Willow Twin) have licensing costs ranging from \$50,000 to \$200,000 annually, completely outside the reach of Peruvian medium-sized companies, whose annual technology budget rarely exceeds \$15,000.

3.3 The Knowledge Gap

There is a significant gap in the academic literature regarding the implementation of predictive Digital Twins in middle-income contexts like Peru, and specifically in small and medium-sized enterprises. Most studies have been conducted in Europe, the United States, or Asia, with budgets and technological capabilities far superior to Peruvian realities (Torres & Méndez, 2025). Additionally, there is limited research on the integration of cognitive learning techniques (such as Case-Based Reasoning) with Digital Twins for construction project management.

3.4 Why This Research Matters (The "So What?" Test)

If this problem is not addressed, Peruvian construction companies will continue to lose competitiveness, public projects will systematically face delays and cost overruns, and the country will not be able to close its infrastructure gap, estimated at over \$160 billion. An accessible predictive model could reduce these losses, democratize access to Construction 4.0 technologies, and serve as a pilot case for other countries in the region with similar realities.

4. RESEARCH QUESTIONS

Selected Paradigm: Design Science (because an artifact is being built: the DT-ML model)

Main Research Question

How can an intelligent digital twin model with cognitive learning, using historical project data, contribute to the prediction and optimization of costs, schedules, and risks in construction projects?

Specific Research Questions

1. How to design a digital twin architecture that integrates historical project data, BIM data, and contextual variables to replicate the dynamic behavior of a construction project?
2. Which cognitive learning techniques (machine learning, deep learning, case-based reasoning) improve the prediction of costs, schedules, and risks in construction projects with limited data volumes?
3. How to structure a digital twin model that enables "what-if" scenario simulation and real-time decision optimization?
4. To what extent does the implementation of the proposed model reduce budgetary, schedule, and risk deviations compared to traditional project management methods (CPM, EVM)?

Hypotheses (for the quantitative evaluation phase)

H1: The Machine Learning model (CatBoost) achieves a Mean Absolute Percentage Error (MAPE) of less than 12% in predicting cost overruns at 50% project completion, significantly outperforming ($p < 0.05$) multiple linear regression (baseline).

H2: The predictive Digital Twin reduces final budget deviation by at least 30% compared to control projects using only traditional methods (CPM and EVM without ML support).

5. OBJECTIVES

5.1 General Objective

To develop an intelligent digital twin model with cognitive learning that, using historical project data, contributes to the prediction and optimization of costs, schedules, and risks in construction projects.

5.2 Specific Objectives (SMART)

#	Objective	SMART Verification
OE1	Design the architecture of a digital twin that integrates BIM data and historical project	S: 4-layer architecture documented; M: Architecture validated by experts; A:



	data to faithfully represent the dynamic behavior of a construction project	Feasible with open-source tools; R: Enables the artifact; T: 6 months
OE2	Identify and apply cognitive learning techniques (ML, CBR) that improve the prediction of costs and schedules in construction projects	S: 5 algorithms compared; M: MAPE < 12%; A: Google Colab resources; R: Core predictive capability; T: 6 months
OE3	Develop a digital twin model that enables simulation of the dynamic behavior of a construction project	S: Functional prototype; M: Runs "what-if" scenarios; A: Open-source tools; R: Core artifact; T: 6 months
OE4	Evaluate the impact of the proposed model in reducing budgetary, schedule, and risk deviations compared to traditional methods	S: Controlled comparison; M: >30% reduction; A: Real projects available; R: Validates utility; T: 8 months

6. METHODOLOGY

6.1 Paradigm and General Method

This research adopts a design science paradigm, as the main objective is to build an artifact (the Digital Twin model with ML engine) that solves a practical problem. Additionally, a quantitative approach is used to evaluate the predictive accuracy of that artifact.

The specific method is a quasi-experiment, as the research works with pre-existing project data (no random assignment of treatments, since projects have already been executed by the companies). This is consistent with the design science paradigm, where the artifact is evaluated by its utility and effectiveness.

6.2 Digital Twin Architecture

The proposed model consists of four layers:

Layer	Description	Technologies	Key Considerations
1. Data Capture	Collection of historical and real-time data from construction projects	Python (pandas, openpyxl), NLP for text extraction	Manual updates + automatic extraction from existing documents
2. Virtual Model	Digital representation of the project (WBS, BIM, or simplified model)	SQLite/PostgreSQL, IFC (if BIM available)	Tolerant to BIM absence; can operate with WBS
3. ML Engine	Training and deployment of cognitive learning algorithms	Python: scikit-learn, XGBoost, CatBoost, TensorFlow	Multiple algorithms compared; SHAP for explainability
4. Visualization	Dashboard with current status, alerts, and scenario simulation	Streamlit or Dash (open source)	Simple, intuitive interface for non-technical users

6.3 Data Sources

Data Type	Source	Estimated Size	Key Variables
Historical projects	3-5 Peruvian construction companies	40-100 projects	Budgeted cost, actual cost, planned duration, actual duration, # change orders, location, soil type, # floors
Current project (validation)	1-2 projects under execution	Weekly monitoring 6-12 months	Physical progress, accumulated costs, incidents
Control projects	Projects using traditional methods	10-20 projects	Final cost/schedule deviation

6.4 Data Preprocessing

Preprocessing steps:

Normalization: Standardization of numerical variables to ensure comparability across projects.

Outlier detection: Use of Tukey's IQR method ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$) and z-score ($|z| > 3$) to identify and treat outliers.

Missing value imputation: K-nearest neighbors imputation; documentation of imputation strategy.

Feature engineering: Creation of derived variables such as complexity ratio, change order density, supervisor experience (coded as junior/semi-senior/senior).

Dataset division:

Training: 70%

Validation: 15%

Testing: 15%

6.5 Algorithms to Be Compared

Priority	Algorithm	Justification
1	CatBoost	Native handling of categorical variables; good performance on small datasets (Hamdan et al., 2025)

2	XGBoost	Industry standard; robust handling of missing values
3	Random Forest	Interpretable; reduces overfitting
4	LSTM Networks	Exploratory; for capturing temporal dependencies
5	Case-Based Reasoning	Cognitive learning; adapts to unique project contexts
6	Multiple Linear Regression	Baseline (minimum to outperform)

6.6 Evaluation Metrics

Metric	Formula	Success Criterion
MAPE (primary)	$(1/n) \sum (\text{actual} - \text{predicted})/\text{actual} \times 100\%$	< 12%
R²	1 - (variance of errors / variance of target)	> 0.75
MAE	$(1/n) \sum \text{actual} - \text{predicted} $	Lower than baseline
RMSE	$\sqrt{[(1/n) \sum (\text{actual} - \text{predicted})^2]}$	Lower than baseline

Validation strategies:

K-fold cross-validation (5 or 10 folds)

Comparison with established baselines

Statistical significance testing (paired t-test, $p < 0.05$)

6.7 Stranger Test Considerations

The project is designed to pass the "stranger test": a person unfamiliar with the project should be able to:

- Clone the repository from GitHub
- Run dvc pull to get the data
- Build the Docker image
- Execute the pipeline and reproduce all results

This requires:

- Complete documentation (README.md, Model Card, Data Sheet)
- Pinned dependencies (requirements.txt)

- Fixed seeds (SEED = 42 for all libraries)
- Clear run instructions

7. ETHICAL CONSIDERATIONS

7.1 Regulatory Compliance

This research complies with:

CONCYTEC National Code of Scientific Integrity (Res. 028-2024): Principles of honesty, transparency, and responsibility

Law No. 29733 (Personal Data Protection Law) : Regulation of personal data processing in Peru

Measure	Description
Anonymization	Removal of company names, professional names, and exact project addresses
Storage	Encrypted UNMSM server; access restricted to researcher and thesis advisor
Informed Consent	Signed document explaining purpose, data use, confidentiality, right to withdraw
Data Sharing	Anonymized data only; raw data not shared without explicit permission

7.3 Risk-Benefit Assessment

Type	Description
Benefits	Participating companies receive free access to the model and training; the construction sector gains an accessible tool
Risks	Minimal (anonymized data); possible perception of inefficiency exposure → mitigated through explicit confidentiality agreements and aggregated results

7.4 Required Approvals

Approval	Entity	Estimated Time
Research Committee	Ethics UNMSM	2-4 months
Data Authorization	Participating construction companies	1-3 months

8. EXPECTED RESULTS

Six specific and verifiable outcomes are expected:

#	Expected Result	Verification Criteria
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1	Reference Architecture: Low-cost Digital Twin architecture	Document validated by peer review; cost < \$15,000
2	Functional Prototype: Predictive Digital Twin model	Public GitHub repository with documentation and tests
3	Structured Dataset: 40-100 Peruvian construction projects	Anonymized dataset available under agreement
4	Algorithm Ranking: Comparison of ML algorithms for Peruvian context	Table with MAPE, R^2 , and statistical significance
5	Deviation Reduction Evidence: Model reduces final budget deviation >30%	Statistical test (t-test) with $p < 0.05$
6	Implementation Guide: Practical guide for Peruvian companies	20-30 page document in Spanish, open license

9. TIMELINE AND BUDGET

9.1 Timeline (3 Academic Years with 30% Buffer)

Period	Activity	Duration (Months)
Year 1	UNMSM Research Ethics Committee approval	3
	Identification and signing of agreements with participating companies	3
	Systematic literature review (PRISMA 2020) and complete theoretical framework	4
	Historical data collection, cleaning, and structuring (40-100 projects)	4
	Year 1 contingency buffer	2
Year 2	ML algorithm training and comparison (6 algorithms, cross-validation)	4
	Digital Twin prototype implementation (4-layer architecture)	4
	Validation on 1-2 ongoing projects (weekly monitoring for 6-8 months)	6
	Year 2 contingency buffer	2
	Year 3	
	Results analysis and statistical hypothesis testing	3
	Thesis writing (complete chapters 1-5)	5
	Preparation of 2-3 journal articles (indexed journals)	3
	Defense preparation (presentation, rehearsal, Q&A)	2
	Year 3 contingency buffer	3
	TOTAL	36

9.2 Budget

Budget Item	Description	Cost (PEN)	Cost (USD)
Hardware & Software			

Cloud computing (Google Colab Pro, 18 months)	For ML model training and experimentation	900	240
Secure server storage (UNMSM - subsidized)	Institutional provision	0	0
Software licenses (open source)	Python, Streamlit, PostgreSQL (free)	0	0
Field Work & Data Collection			
Mobility to participating companies (Lima Metropolitan area)	30 visits × S/50	1,500	400
Audio recorders and transcription software	Otter.ai or similar (annual)	600	160
Printing of informed consent forms and protocols	200 pages	100	27
Personnel & Specialized Services			
Statistical consulting for model validation	3 sessions × S/500	1,500	400
Professional English proofreading for journal articles	2 articles × S/400	800	215
Transcription of interviews (field validation phase)	20 hours × S/30	600	160
Equipment (one-time)			
Laptop upgrade (RAM 32GB + 1TB SSD)	To handle ML model training	2,500	670
External backup drive (2TB encrypted)	For secure data storage	300	80
Dissemination & Publication			
Open access publication fees (1 journal, Q1/Q2)	APC for one article	4,000	1,070
Conference registration (national/international)	1 conference (virtual or hybrid)	800	215
Contingency (20%)	For unexpected expenses	2,400	640
TOTAL		16,300 PEN	\$4,357 USD

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